Irrigation – Benefits and challenges

Australia is a dry continent. However, some parts of Australia, such as far north Queensland, sometimes receive very high rainfall. When this happens, the water flowing along rivers is collected in **reservoirs** and directed to dry areas. This water is used for crops in a process called **irrigation**. Irrigation means that instead of only growing grass and tough shrubs, farmers have enough water to grow citrus and other fruit trees, grass for dairy cattle, grapes, rice and vegetables. Although irrigation helps farmers to grow crops, it also creates some problems.

☞ **1** Match each irrigation problem listed below with its possible solution.

Problem	Possible solution
The open channels and sprinkler systems used for irrigation mean water is wasted due to **evaporation**.	Governments make rules about how much water can be taken from a river.
Some land is too dry for crops to grow, even with irrigation, so the water is wasted	Farmers measure soil moisture to reduce overwatering.
More water is used on crops than is needed.	Mulch and drip systems replace water sprinkling; pipes are used instead of open channels.
Too little water makes it downstream for other uses or for the health of a river.	Measure salinity and plant salt-removing vegetation near irrigated land.
Salt is left behind when water evaporates, causing high **salinity**.	Water authorities only allow irrigation to be used on some crop types.
Too much water is used on crops that don't give a good return.	The federal government buys back extremely dry land from farmers.

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